

Entity Framework (EF & EF Core) - Complete Questions

? Basic Level Questions

1. What is Entity Framework?
2. What is ORM (Object Relational Mapper)?
3. What are the different approaches in Entity Framework?
4. What is Code First, Database First, and Model First?
5. What is the difference between EF and EF Core?
6. What are POCO classes?
7. What is a DbContext class?
8. What is DbSet?
9. How do you configure a connection string in EF Core?
10. What is the use of OnModelCreating() method?

? Code First Related Questions

11. How do you create a Code First model?
12. What is Add-Migration in EF Core?
13. What does Update-Database do?
14. How do you add seed data in EF Core?

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- 15. How do you define relationships (One-to-Many, Many-to-Many)?
- 16. What is Fluent API?
- 17. What is the difference between Fluent API and Data Annotations?

?? Database First / Scaffold Questions

- 18. What is Scaffold-DbContext?
- 19. When do you use Scaffold in EF Core?
- 20. What files are generated when you scaffold a database?
- 21. Can you update your model after changes in the database?

? EF Core Advanced Questions

- 22. What is change tracking?
- 23. What is lazy loading in EF Core?
- 24. What is eager loading? How to use .Include() and .ThenInclude()?
- 25. What is shadow property?
- 26. What is global query filter?
- 27. What is value conversion in EF Core?
- 28. What is Owned Entity?

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29. What is compiled query?

30. How do you write raw SQL in EF Core?

?? Migrations & Versioning

31. What is migration in EF Core?

32. How do you remove or undo a migration?

33. What happens when you change your model after database is created?

34. How do you use migration in a production environment?

? CRUD & Querying

35. How to perform Create, Read, Update, Delete operations in EF Core?

36. How to write LINQ queries with EF?

37. What is AsNoTracking() and when to use it?

38. How to implement soft delete using EF Core?

39. How to update related entities?

40. What is State property in EF Core?

? Performance & Optimization

41. How to improve performance in EF Core?

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42. What are some EF Core best practices?

43. What is the difference between First(), FirstOrDefault(), and SingleOrDefault()?

44. When should you use NoTracking() queries?

45. What is connection pooling?

46. How to log SQL queries in EF Core?

? Transactions & Stored Procedures

47. How to use transactions in EF Core?

48. Can we use stored procedures in EF Core?

49. How do you call a stored procedure and map the result?

50. How to execute raw SQL that returns a non-entity result?

? Testing, Repository Pattern, DI

51. What is InMemory database in EF Core?

52. What is Dependency Injection in EF Core?

53. What is Repository Pattern and how to implement it with EF Core?

54. What is Unit of Work pattern?

?? Real-life Application Scenarios

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55. How would you design a multi-tenant system using EF Core?

56. How to track created/modified timestamps automatically?

57. How do you handle concurrency in EF Core?

58. How do you work with large datasets in EF Core?

? EF Core vs Dapper vs ADO.NET

59. EF vs Dapper ? What's the difference?

60. When to choose EF Core over Dapper?

61. Can EF Core be faster than ADO.NET?

?? Miscellaneous

62. What are navigation properties?

63. What is cascade delete?

64. Can you map a view in EF Core?

65. Can you use EF Core with multiple databases?

66. Can EF Core handle multiple schemas?

67. What is an index in EF Core? How to configure?

68. What are migration bundles?

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69. What's new in EF Core 7 or EF Core 8?

70. How to use logging and diagnostics in EF Core?